

SMART LENDER: LOAN ELIGIBILITY PREDICTION SYSTEM

PROJECT-DESCRIPTION

Smart Lender is a machine learning powered loan eligibility prediction system. It uses Decision Tree, Random Forest, KNN and XGBoost to predict whether an applicant is eligible for a loan based on applicant profile, income, credit history and loan details. The best performing model is deployed in a Flask web application providing real-time predictions, reports, history tracking and batch processing.

TECHNICAL ARCHITECTURE

The Smart Lender system follows a modular machine learning pipeline. Historical loan application data is collected and preprocessed through data cleaning, missing value handling, categorical encoding, and feature transformation. The processed dataset is used to train multiple classification algorithms, including Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost (or Logistic Regression, depending on the selected model). The trained models are evaluated using metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The best-performing model is saved as a serialized model file and integrated into a Flask-based web application. During runtime, applicants submit their loan details through the user interface. The application preprocesses the input using the same pipeline, loads the trained model, predicts loan eligibility, calculates the confidence score, stores the prediction in the history database, and generates downloadable reports. The system also supports batch prediction using CSV files and provides analytical dashboards for monitoring loan approval trends and model performance.

APPLICATION MODULES

- Application: Single applicant prediction.
- Batch Prediction: CSV upload.
- History: Previous predictions.
- Reports: Analytics dashboards.
- Model: Metrics and comparison.

SYSTEM SCREENSHOTS

SmartLender

Underwriting Desk

ApplicationBatchHistoryReportsModel

EnglishDark Mode

APPLICATION

Smart Lender Loan Eligibility Prediction

Premium Underwriting Portal

Enter applicant details to check loan approval likelihood.

Applicant Details

Applicant Name

Property AreaUrban

EducationGraduate

Credit HistoryGood

Prediction Result

Complete the form to see a professional prediction card, approval meter, and loan summary details.

Loan Application

Submit borrower details and let the underwriting engine predict eligibility with confidence.

Applicant Name

Enter name

Gender

Male

Married

Yes

Dependents

0

Education

Graduate

Self Employed

No

AI Loan Assistant

Fill the digital application form to receive an AI-powered loan eligibility prediction with detailed explanations and recommendations.

Paperless Digital Application

AI Loan Eligibility Prediction

Secure Data Processing

Multi-language Support

Explainable AI Prediction

Prediction Confidence Score

Download Prediction Report

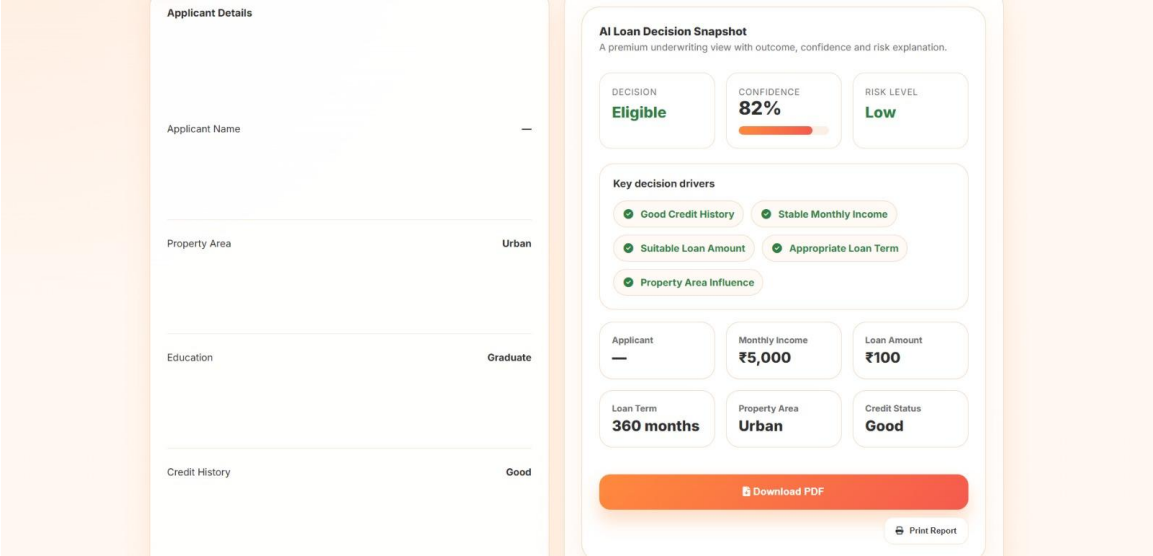
Loan Recommendation

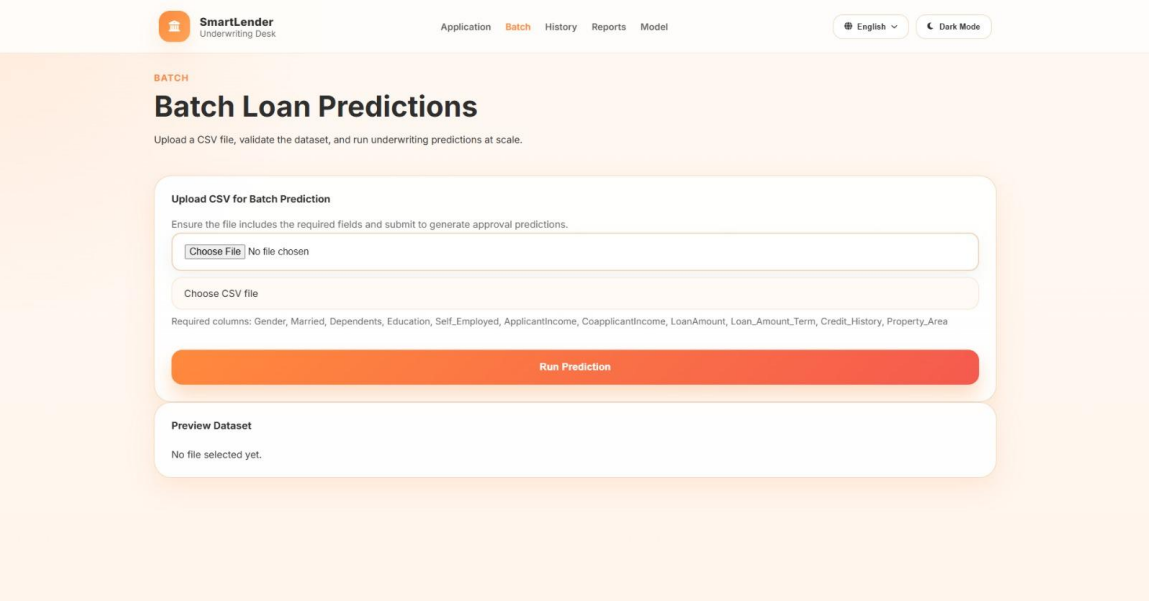
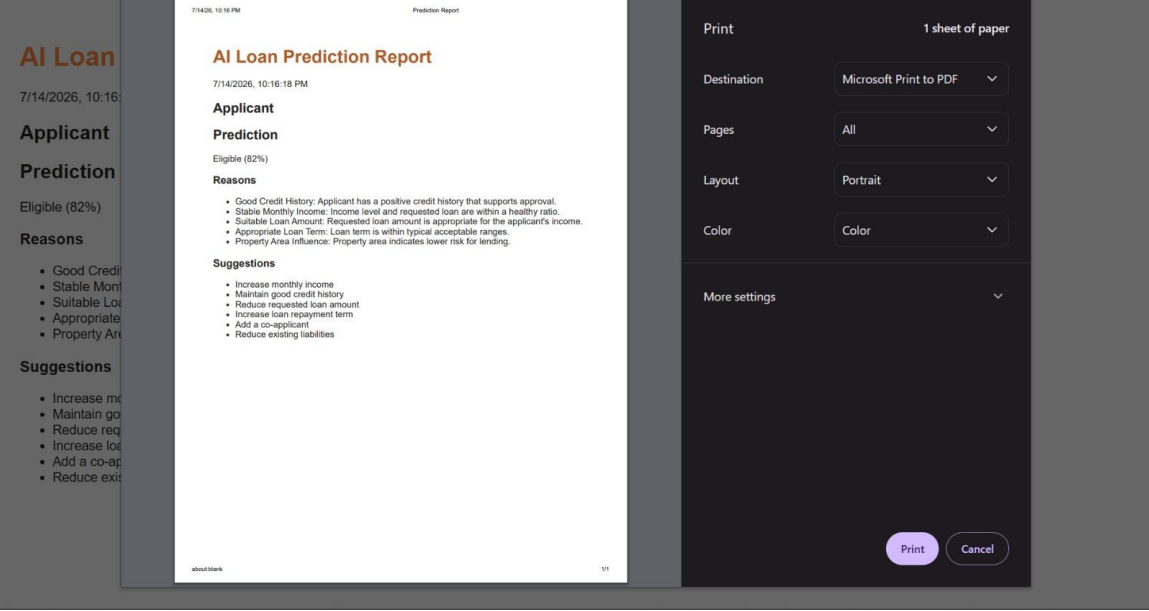
Instant Decision Support

Download Report

Print

New Prediction





SmartLender

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History Records

Prediction ID	Date	Applicant Name	Prediction	Probability	Loan Amount	Income	Status	Actions
APP-20260706140317	2026-07-06 14:03:17	Jane Doe	Eligible	0.82	100	5000	Eligible	—
APP-20260706162343	2026-07-06 16:23:43	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260706170523	2026-07-06 17:05:23	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260706171202	2026-07-06 17:12:02	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260711111923	2026-07-11 11:19:23	Ramu	Eligible	0.79	100	5000	Eligible	—
APP-20260711112525	2026-07-11 11:25:25	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260711112600	2026-07-11 11:26:00	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260711131450	2026-07-11 13:14:50	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260711132622	2026-07-11 13:26:22	chinni	Eligible	0.93	100	500000	Eligible	—
APP-20260711132639	2026-07-11 13:26:39	chinni	Not Eligible	0.0	100000	500000	Not Eligible	—
APP-20260711132651	2026-07-11 13:26:51	chinni	Not Eligible	0.0	100000	500000	Not Eligible	—
APP-20260713071739	2026-07-13 07:17:39	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260713072354	2026-07-13 07:23:54	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260713072620	2026-07-13 07:26:20	N/A	Eligible	0.82	100	5000	Eligible	—
APP-20260713121011	2026-07-13 12:10:11	N/A	Eligible	0.82	100	5000	Eligible	—

SmartLender

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MODEL

Logistic Regression Details

A linear classification model suitable for binary decisions and risk scoring.

Key Metrics

Model NameLogistic Regression
Selected Best ModelLogistic Regression
Training Accuracy0.8004%
Testing Accuracy0.8618%
Cross Validation Accuracy0.7861%
ROC-AUC Score0.8548%
Precision0.84%
Recall0.9882%
F1 Score0.9081%

Algorithm Overview

Logistic Regression models the probability of eligibility using log odds and is interpretable for credit decisions.

Advantages

- Fast training and low complexity.
- Interpretable coefficients.
- Works well with linearly separable data.

Disadvantages

- Can underperform on highly non-linear relationships.

Performance Visuals

REPORTS

Underwriting Analytics

Monitor approval performance and underwriting trends across income, loan amount, and borrower profiles.

Total Applications

0

Approved Loans

0

Rejected Loans

0

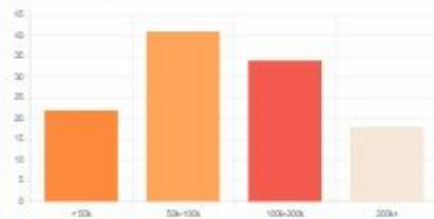
Approval Rate

0%

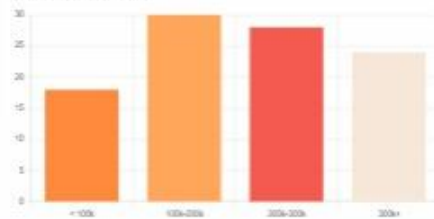
Approval Trend



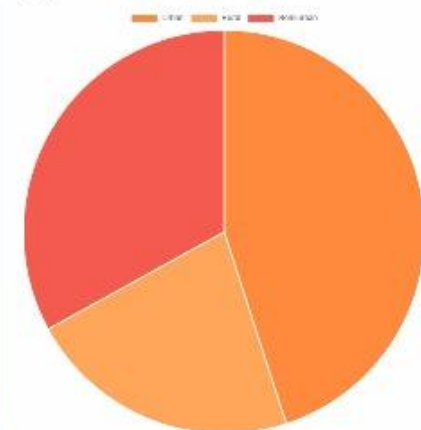
Income Distribution



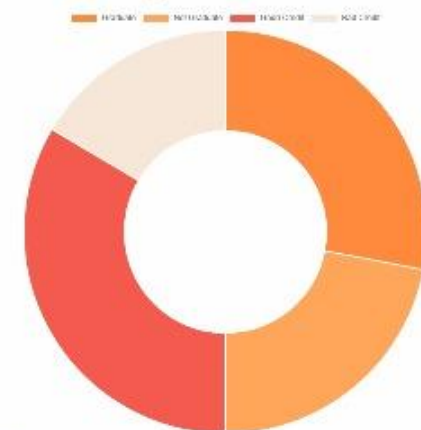
Loan Amount Distribution



Property Area Distribution



Education & Credit History



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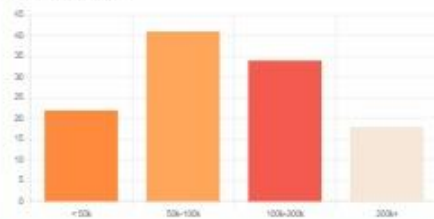
Approval Rate

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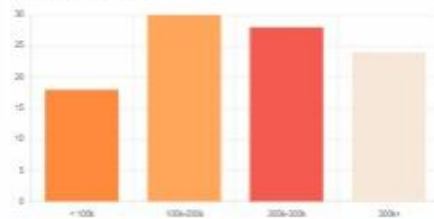
Approval Trend



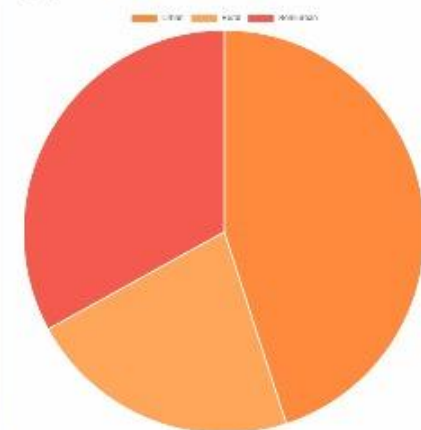
Income Distribution



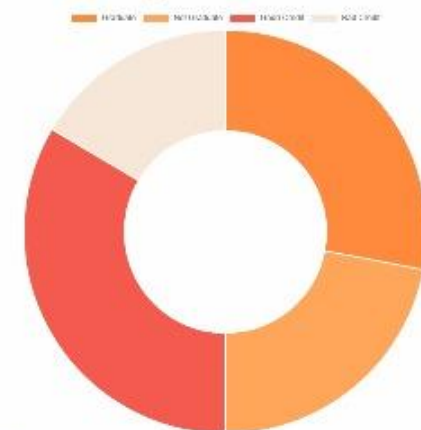
Loan Amount Distribution



Property Area Distribution



Education & Credit History



PRE-REQUISITES

Software	Version	Purpose	Official Link
Python	3.10+	Programming Language	https://python.org
Flask	Latest	Web Framework	https://flask.palletsprojects.com
Scikit-learn	Latest	ML Models	https://scikit-learn.org
XGBoost	Latest	Best Prediction Model	https://xgboost.ai
Pandas	Latest	Data Processing	https://pandas.pydata.org
VS Code	Latest	IDE	https://code.visualstudio.com